

# Capstone Two Proposal - Al Jahnke (Updated)

## Problem to Address

What opportunities exist to develop a short-term stock dividend capture strategy that exceeds the average annual return of the S&P 500 Dividend Aristocrats exchange traded fund (ticker NOBL) by 0.5% after transaction costs?

## Context

Dividend capture is an investment strategy whereby an investor seeks to profit by purchasing a dividend paying stock (preferably one with a high yield) just prior to the stock's ex dividend date in order to earn the upcoming dividend. The investor must be an owner of the stock on the ex dividend date in order to be entitled to the pending dividend, after which the investor can sell the stock in order to avoid further market risk. This strategy is considered low-to-moderate risk but has hidden costs. One such cost is that on the ex dividend date, the stock will typically open lower by the amount of the dividend. The upshot is that while the investor is entitled to the dividend, he/she is immediately faced with a loss of the same amount at the market open. Other costs include transaction costs and market risk (both stock-specific and general market) during holding periods. The strategy will allow for staying in any individual trade if favorable price appreciation is underway.

## Criteria for Success

The construction of a short-term dividend capture strategy that comes the closest to surpassing the average annual return of the NOBL by 0.5% after transaction costs.

## Scope of Solution Space

Starting equity will be \$10,000. Each instance of a dividend capture will have a 4% annual yield minimum. Machine learning models will start on 1/1/2010 and will focus on time series analysis relevant to the following original features:

- Dividend amount
- Ex dividend date
- Daily open
- Daily high
- Daily low
- Daily close
- Stock sector

Engineered features for time series analysis will include the following:

- Daily equity
- Position size
- Dividend yield
- Days to respective upcoming ex dividend date
- Stock beta
- Stock volatility (recent)
- Stock price momentum (recent)
- General market trend
- Sector trend

## Constraints within Solution Space

- Effect of long-term downtrend in general market
- Effect of long-term downtrend in individual market sectors
- Model will only use U.S.-based stocks
- Model will only use companies from the S&P 500 Dividend Aristocrats list

## Stakeholders

Investors with low-to-moderate risk tolerance who are interested in playing for quick-turn passive income and who understand that transaction costs, opportunity costs, and market volatility have the potential to diminish returns.

## Key Data Sources

- Historical data from Yahoo Finance for price, dividends, and sector on the stocks that comprise the S&P 500 Dividend Aristocrats list.
- Historical data from Yahoo Finance for price on the S&P 500 Dividend Aristocrats and State Street sector ETFs.